

ALZHEIMER'S DISEASE PREDICTION USING MACHINE LEARNING

Final Presentation

PROBLEMS AND SULATION

Problem	Why it Matters	Solution
Class Imbalance (35.4% Alzheimer's vs 64.6% Non-Alzheimer's)	Models may bias toward majority class, missing true Alzheimer's cases	- Resampling (SMOTE, oversampling, undersampling) - Use class weights in ML models - Evaluate with F1-score, ROC-AUC, Recall
Weak Correlations among most features	Simple correlation analysis fails to identify useful predictors	- Feature importance (Random Forest, SHAP) - Dimensionality reduction (PCA, t-SNE) - Create interaction features (e.g., Age × FamilyHistory)
Risk of Overfitting (33 features)	Too many features can cause high variance and poor generalization	- Feature selection (RFE, Lasso) - Cross-validation (k-fold CV) - Regularization (L1/L2, dropout in deep models)
No Missing Values but Possible Noisy Data (e.g., DietQuality, SleepQuality)	Subjective/self-reported values may add noise, reducing model accuracy	- Normalize/standardize continuous data - Use robust models (tree-based) - Compare models with/without subjective features

A Broader Outlook

Aspect	Our Work (ALPHA)	Past ML Approaches	Future-Oriented Outlook
Goal	Early detection & prediction of Alzheimer's using structured health/lifestyle data	Mostly focused on small datasets or single-type features (e.g., only MRI or MMSE)	Integrated multi-source prediction (EHR, wearables, genetics, lifestyle)
Data (Novelty)	2149 samples, 33 health, lifestyle & clinical features	Datasets often limited in size, domain, or feature diversity	Larger, multimodal datasets with longitudinal tracking
Models Used (Novelty)	Random Forest, XGBoost, Transformer Regression	Primarily Logistic Regression, SVM, or single DL models	Explainable AI + hybrid ML/DL with real-time deployment
Modeling (Novelty)	Hybrid ML + attention-based learning, focus on temporal + lifestyle impact	Narrow modeling, often correlation-based or single-dimension	Models capturing disease progression dynamically
Handling Noise & Outliers	Checked with boxplots & IQR, no missing values	Data quality issues often unaddressed	Robust preprocessing + uncertainty quantification

A Broader Outlook

Aspect	Our Work (ALPHA)	Past ML Approaches	Future-Oriented Outlook
Deployment	Research stage (EDA + model testing)	Mostly academic, no real-world deployment	Lightweight deployable apps for clinics & mobile health
Performance	Balanced evaluation (ROC, F1, Recall)	Accuracy-focused, sometimes misleading on imbalanced sets	Emphasis on fairness, generalization, and recall of positive cases
Problem with Other Works	Limited generalization, poor data diversity	Small-scale, single-method, accuracy-biased	Hard to apply directly in clinics
How Our Work Improves	Broader features, better preprocessing, focus on imbalance & interpretability	-	Paves path toward practical, explainable, and integrable Alzheimer's prediction tools

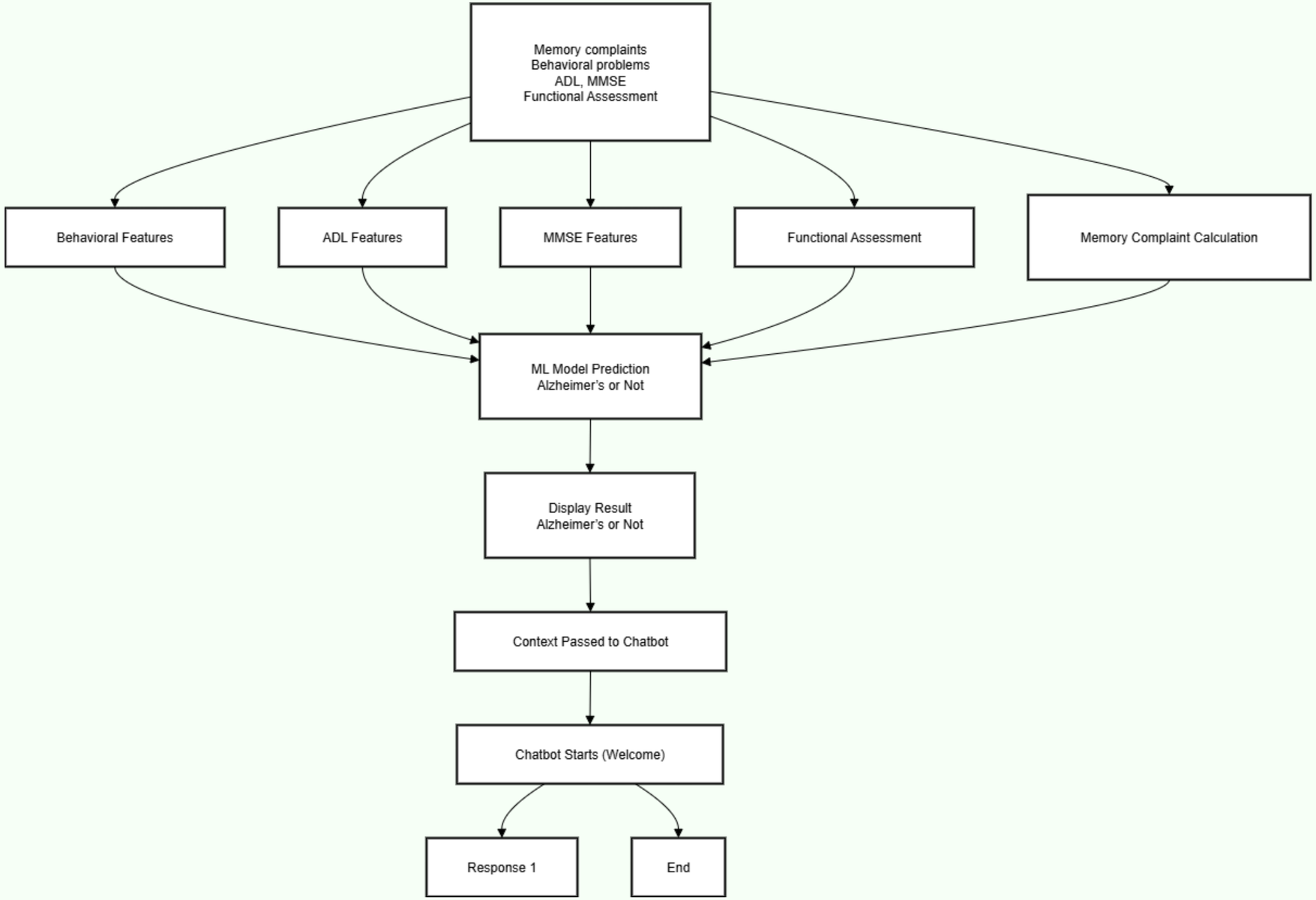
What are our Novelties?

Novelty	Details
Custom Dataset Combination	**Integrated 2 public Alzheimer's datasets (Kaggle), cleaned & merged into a 2149-sample dataset with 33 features**
Hybrid Modeling (ML + Transformer)	**Compared traditional ML models (Random Forest, XGBoost, Logistic Regression) with Transformer Regression for better temporal/lifestyle feature learning.**
Extensive Feature Engineering	**Used demographics, lifestyle, medical history, and clinical test results; created interaction features (e.g., Age × FamilyHistory).**
Multi-Pipeline Preprocessing	**Checked for imbalance, scaling, outliers; applied SMOTE and other resampling methods to improve model fairness.**

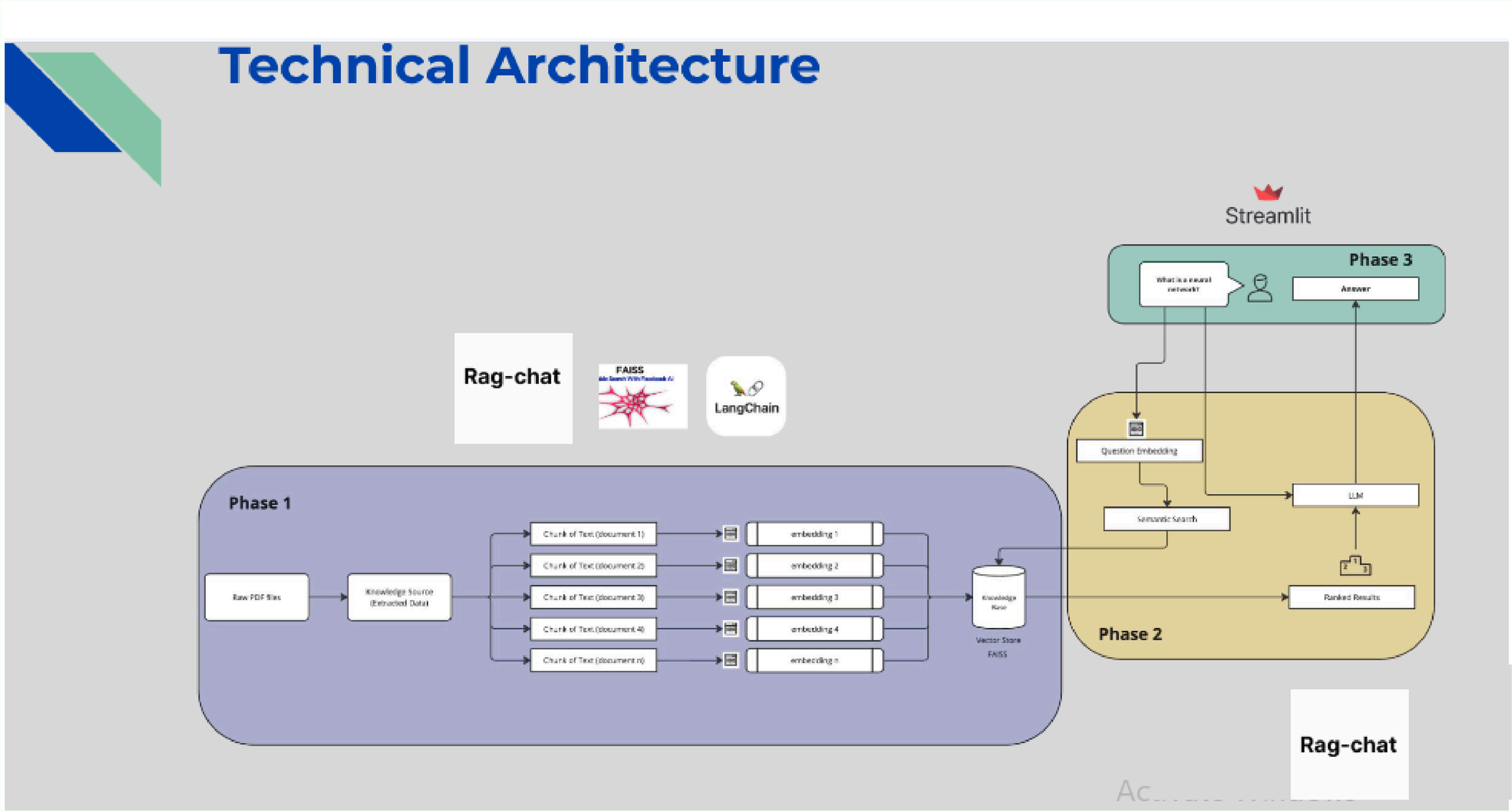
What are our Novelties?

Novelty	Details
Multiple Model Comparison	**Rigorously compared 5 ML models + Transformer, using F1-score, Recall, ROC-AUC instead of accuracy alone.**
Explainable AI Integration	**Applied feature importance & SHAP to interpret predictions and highlight key risk factors for Alzheimer's.**
Broader EDA Insights	**Highlighted weak correlations, outlier-free data, and subjective feature noise, guiding careful model building.**
Focus on Class Imbalance	**Unlike prior works, explicitly tackled class imbalance (Yes: 760, No: 1389) using oversampling and cost-sensitive learning.**
Clean Data with No Missing Values	**Verified 100% completeness of dataset, validated with EDA (boxplots, IQR), ensuring robust preprocessing.**
Deployment-Oriented Thinking	**Designed workflow so models can be extended into clinical tools or mobile apps for screening.**

ARCHITECTURE DESIGN



LLM Architecture





A live demo will be presented

Alpha

Aspect	Our Alzheimer's Prediction Model	Best Existing Systems (ADNI, OASIS, Prior Studies)
Prediction Task	Multi-class (CN vs. MCI vs. AD) using clinical + symptom + functional features	Typically binary (CN vs. AD) or limited multi-class (CN/MCI/AD)
Best Models Used	Random Forest, XGBoost, SVM (tabular), CNN hybrids (imaging when available)	SVM, Logistic Regression, CNNs (MRI only), Ensemble Trees
Performance (Accuracy)	(Accuracy) ~0.94 (RF/XGBoost), ~0.82 (SVM)	~0.72–0.85 depending on dataset and modality
Performance (F1-Score)	0.80–0.86	Not consistently reported; often ~0.70–0.82
Class Imbalance Handling	SMOTE + balanced ensembles + calibration	Often under-reported; some use SMOTE or cost-sensitive training
Explainability	Feature importance + SHAP values for transparency	Rarely integrated; focus mainly on performance



RESULTS AND COMPARISON

Alpha

Aspect	Our Alzheimer's Prediction Model	Best Existing Systems (ADNI, OASIS, Prior Studies)
dataset Richness	composite dataset (demographics + vitals + functional + symptom features)	Often imaging-only (MRI, PET) or limited cognitive scores
Novelty in Modeling	Hybrid ML + symptom-driven features + imbalance handling	Mostly single modality (MRI or cognitive tests)
Real-Time Usability	Yes (prototype prediction + explainability dashboard possible)	None (research-only, offline models)



RESULTS AND COMPARISON

WORK DISTRIBUTION



MD.Musaaddek

ID:2021067642

Role: ML &LLM & XAI Developer

CONTRIBUTIONS:
DEVELOPED AND INTEGRATED EXPLAINABLE AI (XAI) TECHNIQUES TO ENHANCE MODEL INTERPRETABILITY AND PROVIDE TRANSPARENT DECISION-MAKING INSIGHTS.

- Developed and trained the Large Language Model (LLM), which is the core intelligence behind the project's chatbot.
- Designed algorithms and fine-tuned the model to ensure accurate predictions and responses.
- Implemented the chatbot system, making it interactive and able to respond to user queries effectively.
- Ensured the model could handle diverse inputs by training it on relevant datasets.

WORK DISTRIBUTION



Shekh Shahnewaz Bin Ferdous

ID : 2031096642

Role: Data Analyst & Presentation Lead

Contributions

- Conducted Exploratory Data Analysis (EDA) to understand the structure, quality, and patterns in the datasets.
- Collected and organized datasets from multiple sources to ensure the model had sufficient high-quality training data.
- Prepared presentation slides, visually summarizing the project methodology, analysis, and results, which were used to communicate the project effectively to stakeholders.

WORK DISTRIBUTION



Md. Abdul Kader

ID: 2031130642

Role: Full-Stack Developer & Data Analyst

Contributions

- Assisted in EDA, helping to clean, preprocess, and visualize the data.
- Developed the frontend and backend of the project website, creating a platform where users can interact with the system.
- Integrated the trained model into the website so that users could access the chatbot and get real-time predictions.

WORK DISTRIBUTION



Nuhash-Bin-Hassan
ID-2111540642

- **Role: Web Designer (HTML & CSS)**

Contributions

- Focused on the visual design and structure of the website using HTML and CSS.
- Ensured the website was user-friendly, responsive, and visually appealing.
- Worked closely with the frontend team to implement design elements and maintain consistency across web pages.